ALICE SOMIGLIANA

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Italian

EMPLOYMENT AND RESEARCH EXPERIENCE

Max Planck Institute for Astronomy, Heidelberg (DE) *2025-ongoing*
Postdoctoral Researcher | Planet Formation and Exoplanets (PFE) Department
Advisors: Giulia Perotti, Myriam Benisty (MPIA)

EDUCATION

PhD in Physics *2021-2024*
European Southern Observatory (ESO), Garching bei München (DE)
Awarding institute: Ludwig-Maximilians-Universität München (LMU), magna cum laude (1.0)
Supervisor: Prof. Leonardo Testi

Masters in Physics *2019-2021*
University of Milan, 110/110 cum laude
Thesis: *A population synthesis model for young stars and their disks*
Supervisor: Prof. Giuseppe Lodato

Bachelors in Physics *2015-2018*
University of Milan, 105/110
Thesis: *Evolution of protoplanetary disks undergoing photoevaporation*
Supervisor: Prof. Giuseppe Lodato

RESEARCH INTERESTS

Secular evolution of protoplanetary disks and its impact on planet formation and disk observables, approached from the theoretical and numerical perspective. Population synthesis modelling, including dedicated software development. Disc chemistry, JWST/MIRI-MRS observations.

FIRST-AUTHOR PEER-REVIEWED PUBLICATIONS

Somigliana A., Testi L., Rosotti G., Toci C., Lodato G., Anania R., Tabone B., Tazzari M., Klessen R., Lebreuilly U., Hennebelle P. and Molinari S. (2024), *The evolution of the $M_d - M_\star$ and $\dot{M} - M_\star$ correlations traces protoplanetary disc dispersal*, A&A, 689, A285. Citations: 4

Somigliana A., Testi L., Rosotti G., Toci C., Lodato G., Tabone B., Manara C. F. and Tazzari M. (2023), *The Time Evolution of $M_d/\dot{M}$ in Protoplanetary Disks as a Way to Disentangle between Viscosity and MHD Winds*, ApJL, 954, L13. Citations: 16

Somigliana A., Toci C., Rosotti G., Lodato G., Tazzari M., Manara C. F., Testi L., and Lepri F. (2022), *On the time evolution of the $M_d - M_\star$ and $\dot{M} - M_\star$ correlations for protoplanetary discs: the viscous time-scale increases with stellar mass*, MNRAS, 514, 5927-5940. Citations: 20

Somigliana A., Toci C., Lodato G., Rosotti G. and Manara C. F. (2020), *Effects of photoevaporation on protoplanetary disc 'isochrones'*, MNRAS, 492, 1120-1126. Citations: 29

CO-AUTHOR PEER-REVIEWED PUBLICATIONS

Kurtovic N., et al., including **Somigliana A.** (2026), *MINDS: Young binary systems with JWST/MIRI: Variable water-rich primaries and extended emission*. A&A, 705, A97.

Tabone B., et al., including **Somigliana A.** (2025), *The ALMA survey of Gas Evolution in PROtoplanetary disks (AGE-PRO): VII. Testing accretion mechanisms from disk population synthesis*. ApJ, 989, 7. Contribution: main software developer, software user support.

Malanga L., Rosotti G., Lodato G., **Somigliana A.**, et al. (2025), *The survivorship bias of protoplanetary disc populations. Internal photoevaporation increases the median total disc mass with time.* A&A, 699, A292. Contribution: co-supervisor of Masters' student (Malanga L.), main software developer.

Zallio L., et al., including **Somigliana A.** (2024), *On the accretion of MHD wind-driven proto-planetary discs: the $M_D - \dot{M}_*$ relation*. A&A, 692, A93.

Lebreuilly U., et al., including **Somigliana A.** (2024), *Synthetic populations of protoplanetary disks - Impact of magnetic fields and radiative transfer*. A&A, 682, A30.

ACCEPTED OBSERVING PROPOSALS

JWST

On the shoulders of a giant. Towards a chemical inventory of Herbig discs. Cycle 5. 16.9 h. **PI: Alice Somigliana**

ALMA

The missing piece of disc evolution: disc demographics in 25 Orionis, an old region with low UV radiation. Cycle 10. 15.2 h. PI: Francesco Zagaria

An ALMA Band 1 and VLA survey to probe the solid reservoir of Lupus disks. Cycle 10. 28.6 h. PI: Laura Perez

NOEMA

Protoplanetary discs at the end of their life-time. 2022. 6 h. PI: Benoît Tabone

SOFTWARE DEVELOPMENT

Diskpop and popcorn

Main active developer. `Diskpop` is a Python tool to perform population synthesis of protoplanetary disks, whose raw output is analysed with the `popcorn` library. I wrote the documentation and tutorials to both codes, released in Somigliana et al. 2024.

TALKS

SPiCE 2, Lyon. Invitation-only conference.	*March 2026*
SPF Seminar, ESO, Garching.	*March 2026*
Lunch Talk, Leiden.	*February 2026*
ExoCam Seminar, Cambridge. **Invited seminar**	*January 2026*
Italian National Conference of Star and Planet Formation, Sexten. **Invited talk**	*January 2025*
New Heights in Planet Formation, ESO, Garching. Contributed talk	*July 2024*
OPINAS Seminar, MPE, Garching.	*July 2024*
EAS 2024, Padova. Two contributed talks	*July 2024*
SESTAS Seminar, MPA, Garching.	*January 2024*
Milan XMas Meeting, Milan. Contributed talk	*December 2023*
Core2diskIII, Paris. Invitation-only conference.	*October 2023*

TOP Seminar, Observatoire de la Cote d'Azur, Nice.	*September 2023*
PSF Seminar, MPIA, Heidelberg.	*July 2023*
EAS 2023, Krakow. Contributed talk	*July 2023*
AGE-PRO workshop, Leiden.	*January 2023*
Disks and Planets across the ESO Facilities. Contributed talk	*November 2022*
Arcetri Observatory Seminar, Florence.	*June 2022*
IMPRS Symposium, MPE, Garching. Contributed talk	*June 2022*
Milan XMas Meeting, Milan. Contributed talk	*December 2021*
ECOGAL Seminar, online	*September 2021*
Milan Xmas Meeting, Milan. Contributed talk	*December 2019*

SUPERVISION

Gloria Giamberardini — *2026*
Erasmus Traineeship student at MPIA.

Jasmine Khalil — *2025*
Summer Student at MPIA. Co-supervised with Francesco Zagaria (MPIA).

Lorenzo Malanga — *2024*
Master's student at the University of Milan. Co-supervising with Giovanni Rosotti and Giuseppe Lodato (University of Milan). Results in Malanga et al. (2025). Currently PhD Student at the University of Milan.

Daria Zaremba — *2023*
Summer Student at ESO. Co-supervised with Marta de Simone and Łukasz Tychoniec (ESO). Results in Giani et al. (in prep).

GRANTS AND AWARDS

ESO Science Internship — *2023*
Awarded to supervisors to support Master's students visits at ESO. I obtained funding to allow Lorenzo Malanga to visit ESO for three months (March - June 2024) to work with me on his thesis. Total value ~ 4500 €.

SOC AND LOC EXPERIENCE

DUSTBUSTERS New Heights in Planet Formation — *2024*
LOC member

EAS 2024 — *2023*
SOC Member of the proposed Special Session *Young protostellar disks: the initial conditions for planet formation.*

Disks and Planets across the ESO Facilities — *2022*
LOC member

IMPRS Symposium — *2022*
LOC member

COMMUNITY SERVICE AND OUTREACH

MPIA PFE Coffee Organiser — *2025-ongoing*
Topical weekly seminar at MPIA. Responsibilities include finding speakers, managing communications and reminders.

ESO Journal Club Organiser *2023-2024*
Weekly meeting gathering the whole ESO Office for Science. Responsibilities include finding speakers, managing communications and reminders, chairing, acquiring refreshments.

Student Representative *2022-2024*
Currently in charge of communication with the Office for Science (2023-2024), previously welcoming new students and providing help with bureaucracy (2022-2023).

SPF Coffee Organiser *2021-2024*
Topical weekly coffee at ESO

OPC Scientific Assistant *Cycle 112, 113*
Assisting panelists reviewing ESO proposals

Volunteer at ESO Supernova *2021-2024*
Exhibition, Planetarium, and ESO HQ tours in italian for high school students.

LANGUAGE SKILLS

Italian (Native speaker), English (Proficient), German (B1), French (Basic)

REFERENCES

Prof. Leonardo Testi
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Prof. Richard Alexander
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